HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and implemented automated product testing procedures, resulting in a **15% increase** in test coverage for new hardware releases.
- Created, tested, and maintained **factory test software** for core product lines, reducing pre-production defect rates by **~8%**.
- Executed data analysis on **test results and parametric data**, identifying failure trends that informed **hardware design adjustments** and improved yield.
- Collaborated with **Test Engineering, HW, and SW teams** globally to integrate new testing methodologies into existing product lifecycles.
- Provided **technical direction and support** to technician and production support teams, resolving an average of **5 critical issues** per week during peak production.
- Authored and reviewed comprehensive **technical documentation** for test processes and software, ensuring clear release and adoption guidelines for global teams.
- Containerised test execution environments using **Docker**, enabling consistent and repeatable deployments across distributed testing facilities on **AWS EC2**.

SKILLS

Testing – Automated Product Testing, Factory Test Software, Test Process Development, Test Results Analysis, Parametric Data Analysis, Hardware Design Adjustments, Yield Improvement

Programming & Scripting – Python, Java, SQL, Bash Scripting, REST APIs

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD

Data Analysis & BI – MS Excel, Statistical Analysis, Data Visualization, Power BI, Tableau, DAX

Operations & Support – Product Operations, Manufacturing, New Product Introduction, Production Support, Documentation, Change Management, Incident Response

Project Management – Agile Methodologies, Scrum, Requirements Gathering, Stakeholder Communication

PROJECTS

Automated Test Suite Development for Embedded Systems

- Engineered a Python-based test automation framework to validate firmware functionality on embedded devices, achieving **92% test case pass rate**.
- Integrated **pytest** and **Jenkins** to create a CI/CD pipeline for automated firmware testing, reducing manual testing effort by **~40%**.
- Developed REST APIs using **Flask** to control test hardware and collect real-time diagnostic data, enabling faster debugging cycles.

Manufacturing Process Optimization Dashboard

- Designed and developed an interactive **Power BI** dashboard to visualize key manufacturing metrics including **Yield, Units Per Hour, and Order Cycle Time** for an enterprise client.
- Implemented complex **DAX measures** and **Power Query** transformations to aggregate and clean data from disparate factory systems.
- Collaborated with operations stakeholders to refine dashboard requirements, directly impacting decision-making for process improvements.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)